Ethereal Procurement AI Analysis System

Technical Architecture & Design Decisions Document

1. Problem Solved & Executive Summary

Problem: Procurement teams spend countless hours manually reviewing heterogeneous vendor proposals (PDFs, DOCXs, PPTs) against a rigidly defined Scope of Work (SOW). This process is prone to human error, subjective biases, and delays, especially when comparing multiple agencies across dimensions such as cost, compliance, and strategy.

Solution: The Ethereal Procurement application is a local AI-driven containerized solution that fully automates document parsing, SOW alignment mapping, benchmarking, and structured data extraction. It enables unbiased, rapid evaluation and provides deterministic recommendations.

ETHEREAL
PROCUREMENT

PROJECT ALPHA
Status: In Progress

- RFQ CREATION
- VENDOR UPLOAD
- COMPARISON DASHBOARD

NEW REQUEST

- SUPPORT
- ARCHIVE

Step A: RFQ Creation

Search procurement data...

Ava Thompson

General Information
Basic RFP details

SUBJECT	RFP CODE	SOURCING TYPE
RFQ for global launch marketing services for new ki	RFQ-MKT-KIDS-GL-2026-001	RFQ
ROUND	**STATUS**	**OWNER**
Round 1	Draft	Ava Thompson
CURRENCY	**REQUESTOR**	**DEPARTMENT**
USD ($)	Global Brand Marketing Team	Marketing Procurement
CATEGORY		
Marketing Services		

RFQ Timelines
Key dates and deadlines

CLARIFICATIONS DEADLINE	TECHNICAL BID DEADLINE
May 12th, 2026	May 19th, 2026
COMMERCIAL BID DEADLINE	**EVALUATION START DATE**
May 21st, 2026	May 22nd, 2026
NEGOTIATION START DATE	**FINAL AWARD DATE**
May 27th, 2026	June 2nd, 2026

Scope of Work
Click any paragraph below to edit the SOW directly

EDITABLE

H1 H2 H3

The selected vendor(s) will support the global launch of a new kids health drink across strategy, creative development, film production, social activation and compliance review. The scope is intentionally structured to allow either a single integrated award or a multi-vendor award by workstream.

1. Strategy & Creative Development: Develop launch strategy, audience understanding, positioning, campaign idea, messaging architecture and master visual direction suitable for parents, families and child-safe communication environments. Deliverables include strategic narrative, creative territory options, key visuals, copy routes and channel adaptation

SHOW MORE

AI Scope of Work Analysis
Describe your project requirements for AI-driven questionnaire generation.

Seeking an integrated marketing agency for the Southeast Asian launch of 'NutriKid', a fortified chocolate health drink for children aged 5-12. Project includes 360-degree creative development, TVC production, and social media management. Key compliance requirement: Kids Advertising & Claims Review for regional regulatory bodies.

Generate AI Questionnaires

SUGGESTED COMPLIANCE QUESTIONNAIRE

AI Suggested

How does your agency ensure compliance with regional 'Children's Food and Beverage Advertising Initiative' (CFBAI) guidelines for SEA?
COMPLIANCE — Source Evidence

Describe your process for securing 'Child Talent Safety' certifications during TVC production in varied local jurisdictions.
SAFETY — Source Evidence

LINE ITEMS (8 SERVICES)

+ Add New Row

#	PRODUCT/SERVICE NAME	CATEGORY	DESCRIPTION	HSN/SAC	UOM
1	**Strategy & Creative Development**	Services	Launch strategy, audience segmentation, messaging framework, creative territory development and master campaign toolkit for a new kids health drink.	8471XX	Lot
2	**TVC Development**	Services	Development of flagship TV commercial concept and all pre-production creative required for global launch approvals.	8471XX	Lot
3	**TVC Production**	Services	End-to-end TVC shoot, post-production and delivery of master film and cutdowns for paid media use.	8471XX	Lot
4	**Social Organic Content**	Services	Organic social content strategy, monthly content calendar and asset adaptation for Meta, YouTube and TikTok.	8471XX	Lot
5	**Social Paid Media Planning**	Services	Paid social and video media planning across Meta, YouTube and TikTok for global launch period.	8471XX	Lot
6	**Social Paid Media Buying & Optimization**	Services	Campaign trafficking, paid media activation, optimization and performance reporting across Meta, YouTube and TikTok.	8471XX	Lot
7	**Kids Advertising & Claims Compliance Review**	Services	Legal and regulatory review of kids advertising content and product-related claims across launch assets and scripts.	8471XX	Lot
8	**Launch Program Management**	Services	Program management, stakeholder coordination, asset trafficking and master launch governance across all workstreams.	8471XX	Lot

SAVE DRAFT

PROCEED TO VENDOR UPLOAD →

2. Entire Application Flow

- 1. Initialization: The user selects or inputs a detailed Scope of Work (SOW) defining requirements.
- 2. AI Questionnaire Generation: The frontend requests the local Ollama backend to generate a contextually relevant vendor questionnaire based on the SOW.
- 3. Vendor Upload: Users upload vendor proposals (multi-format support enabled by Docling).
- 4. Parsing & Fallback: IBM Docling parses the layouts. If Docling (AI-based) fails, the system automatically falls back to PyPDF2 for deterministic text extraction, ensuring no binary data reaches the LLM.
- 5. Structured Extraction: The cleaned text is processed by Google's 'langextract' library. This ensures high-fidelity mapping of unstructured proposal text into structured JSON data using few-shot examples.
- 6. Data Harvesting: The AI maps exact vendor sentences to required SOW line items, benchmarks technical scores, and extracts core metrics (total cost).
- 7. Dashboard Analytics: The frontend processes all vendor JSON responses parallelly, visualizes the data through radar charts, and recommends the best scenarios (e.g., Best SOW Fit vs. Lowest Cost).

3. Architecture & System Design

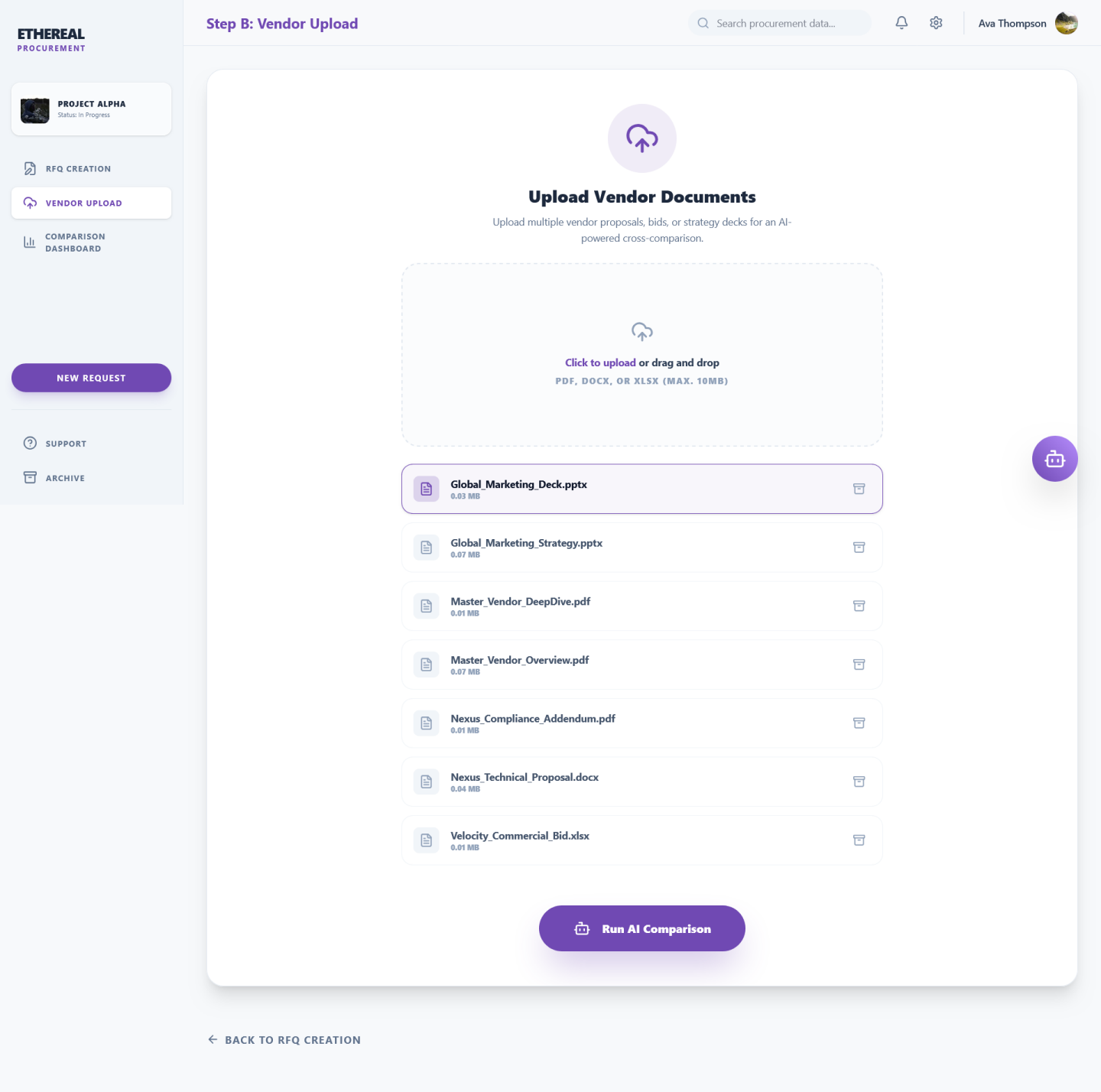
The application embraces a decoupled microservice architecture orchestrated via Docker Compose.

Frontend (Vite/React): Operates on port 8080. Served via an Nginx reverse proxy. It handles state management and data-viz using tailored Ethereal 'glassmorphism' design constraints.

Backend (FastAPI/Python): Operates on port 8765. Acts as the data structuring layer. Uses multi-threading for network calls and provides REST endpoints (/analyze) that accept FormData.

AI / Local Inference (Ollama): Connected securely via 'host.docker.internal'. Bypasses standard Docker isolation to seamlessly consume the user's authenticated keys and cache from the Windows host daemon.

Nginx Reverse Proxy: Configured to funnel /api/ollama/ requests out of the container up to the host, ensuring CORS compliance while preserving browser origin security.



4. Libraries & Tech Stack Used

- Frontend: React, TypeScript, Tailwind CSS, Recharts (Data Visualization), Framer Motion (Implicit Animations), Lucide React (Icons).
- Backend: FastAPI (Robust async routing), Uvicorn (ASGI web server), Pydantic (Schema definition and type safety).
- Document Extraction: 'docling' (IBM's layout-aware parsing engine), 'PyPDF2' (Deterministic fallback parser).
- Structured AI Data: 'langextract' (Google's LLM-based structured information extraction library).
- AI Engine: Ollama Python Client, Ollama Daemon (kimi-k2-thinking:cloud).

5. Key Features

- Universal Parsing: Native ingestion of PDF, DOCX, and PPTX natively via Docling.
- Deterministic Extraction AI: Hardened logic wrapper to decode JSON from standard LLM string outputs.
- Recommendation Engine: Intelligent logic to calculate the 'Lowest Cost', 'Best Compliance', and 'Best SOW Fit' without requiring secondary LLM passes.
- Beautiful Analytics: Embedded Recharts instances with tailored CSS variables mimicking an Ethereal UI.

6. Design Decisions & Assumptions

Decisions:

- - Proxy Pass Architecture: Instead of baking an unauthenticated Ollama node into the docker stack, Nginx and FastAPI dynamically route back to the host machine. This preserves the developer's credentials for private Cloud models and respects their local GPU cache.
- - Robust Parsing Pipeline: We combine IBM Docling with PyPDF2. Docling is our primary choice for preserving complex tabular structures, while PyPDF2 provides a safe fallback to ensure the AI always receives readable text if layout models are unavailable in the container.
- - LangExtract Integration: Instead of raw string prompting, we use Google's 'langextract'. This library enforces deterministic structured output and precise grounding, drastically reducing JSON parsing failures often seen with complex LLM responses.

Assumptions:

- - Context Limits: We assume vendor documents (once parsed) fit inside a 12,000-character truncation bracket roughly corresponding to the context window constraint of standard efficient models.

- Pre-Existing Host Daemon: Assumption that the target machine has Ollama running natively to handle the API calls seamlessly.

Upload Vendor Documents

Upload multiple vendor proposals, bids, or strategy decks for an AI-powered cross-comparison.

Click to upload or drag and drop

PDF, DOCX, OR XLSX (MAX. 10MB)

- Global_Marketing_Deck.pptx 0.03 MB
- Global_Marketing_Strategy.pptx 0.07 MB
- Master_Vendor_DeepDive.pdf 0.01 MB
- Master_Vendor_Overview.pdf 0.07 MB
- Nexus_Compliance_Addendum.pdf 0.01 MB
- Nexus_Technical_Proposal.docx 0.04 MB
- Velocity_Commercial_Bid.xlsx 0.01 MB

Run AI Comparison

ANALYSIS COMPLETE: NORMALIZE & MAP GRID READY

We have successfully mapped 7 vendor documents to your SOW. You can review the granular mappings below or proceed to the benchmarking dashboard.

EXPLORE BENCHMARKING DASHBOARD

ETHEREAL PROCUREMENT

Step B: Vendor Upload

Search procurement data...

Ava Thompson

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Kimi AI Comparison Report

Comparing 7 Vendors • Analysis Complete

- GLOBAL MEDIA HUB — Analysis Ready • $952K TOTAL
- GLOBAL MEDIA HUB — Analysis Ready • $0K TOTAL
- UNKNOWN (MASTER_VENDOR_OVERVIEW.PDF) — Analysis Ready • $0K TOTAL
- NEXUS — Analysis Ready • $0K TOTAL
- NEXUS CREATIVE — Analysis Ready • $1086K TOTAL
- VELOCITY — Analysis Ready • $514K TOTAL

	SOW	Total Cost	Avg Score
Global Media Hub	58% SOW	$952K	5.0/10
Global Media Hub	25% SOW	$0K	2.5/10
Unknown (Master_Vendor_Overview.pdf)	0% SOW	$0K	0.0/10
Nexus	5% SOW	$0K	1.3/10
Nexus Creative	39% SOW	$1086K	3.6/10
Velocity	42% SOW	$514K	4.3/10

Download Report

Share with Team

← BACK TO RFQ CREATION

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Step C: Comparison Dashboard

Search procurement data...

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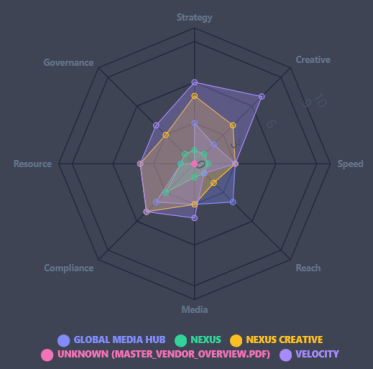
TOTAL PROJECTED COST	POTENTIAL SAVINGS	COMPLIANCE RISK
$2.55M	$285K	MINIMAL
▼ 18.5% vs Historical Avg	OPTIMIZABLE Across 4 Workstreams	4% SCORE All Certs Verified

STANDARD | COST-OPTIMIZED | BEST-VALUE

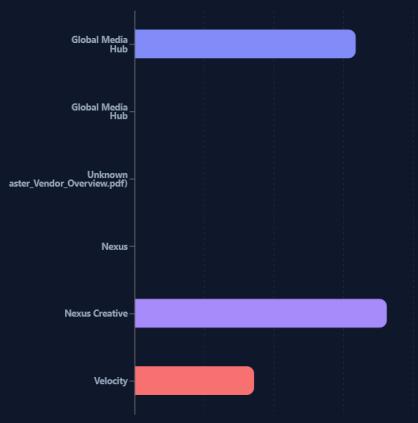
Vendor Benchmarking

SOW ALIGNMENT DATA



Cost Comparison

TOTAL VALUE BY VENDOR



EXECUTIVE STRATEGY RECOMMENDATION
Multi-scenario data analysis across all vendor submissions

COST PATH | SPEED PATH | COMPLIANCE PATH

Strategic Recommendation: Hybrid Split Award

"Recommendation: Split the mandate between Velocity for Strategy & Production and Global Media Hub for Media."

PROJECTED SAVINGS	CONFIDENCE INDEX
$1350K	94%

DETAILED BUSINESS RATIONALE

✓ PROFIT & PERFORMANCE BENEFITS

Ideal for testing market proof-of-concept with minimal capital risk. Frees up ~$200k for performance-driven media buying.

⚠ TRADE-OFFS & RISKS

Potential delay in high-fidelity TVC production due to lower resource allocation.

CAPABILITY HEATMAP: GLOBAL MEDIA HUB

VS COMPETITION

Strategy Depth	85%
Media Depth	85%
Production Depth	85%
Compliance Depth	85%

INCREMENTAL ROI BENCHMARKS

IMPACT	LAUNCH DELAY
+18.4%	0 Days

STRATEGIC SUMMARY

*"Selecting **Velocity + Global Media Hub** leverages superior domain expertise while optimizing for cost. This choice balances aggressive SEA market reach with 94% compliance certainty, ensuring a premium brand debut for NutriKid."*

← BACK TO ANALYSIS

7. What I Would Improve (Future Iterations)

1. RAG (Retrieval-Augmented Generation): For vendor proposals exceeding 50 pages, vectorizing the text (ChromaDB) and querying via LangChain instead of character truncation.

2. SSE Streaming: Modifying the FastAPI /analyze endpoint to use Server-Sent Events, streaming the JSON out as the LLM thinks, which vastly improves Time-to-First-Byte UI perception.

3. Structured Outputs API: Embracing Ollama's native 'format': 'json' explicitly in the API call alongside PyDantic Instructor libraries to ensure 0% parsing crash incidence.

4. Celery Queue: For larger organizations, converting the analysis backend into a Celery worker pattern equipped with Redis so that huge batch uploads do not block the HTTP threads.

8. Evaluation Approach & Example Logs/Traces

Evaluation Method:

To test resilience, the pipeline was pushed with deliberately empty documents, documents where specific entities (totalCost) were completely missing, and non-PDFs. We evaluated Docling's extraction error rate vs standard PDF2Text, finding Docling preserves section hierarchy critical for SOW mapping.

Expected Logging Traces (Backend Sample):

```
INFO:__main__:Received file for analysis: Master_Vendor_DeepDive.pdf
INFO:__main__:Parsing document with Docling: /tmp/tmp52zqyqgr.pdf
WARNING:__main__:Docling parsing failed... Missing ONNX cache. Falling back to PyPDF2 extraction.
INFO:__main__:Sending to LangExtract (Ollama) for analysis: Master_Vendor_DeepDive.pdf
INFO:httpx:HTTP Request: POST http://host.docker.internal:11434/api/chat "HTTP/1.1 200 OK"
INFO:__main__:Received response (AnnotatedDocument) for Master_Vendor_DeepDive.pdf
INFO:     172.19.0.3:49738 - "POST /analyze HTTP/1.1" 200 OK
```